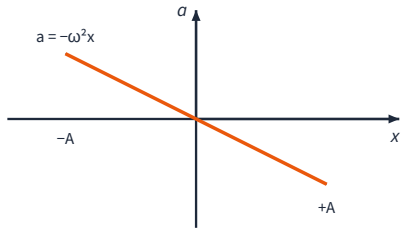
CHAPTER 19

# OSCILLATIONS

**Key idea:** in SHM the acceleration is proportional to the displacement and directed towards equilibrium.

## 1 SIMPLE HARMONIC MOTION




$$a = -\omega^2 x \qquad F = -kx$$

- $a \propto x$, always towards the equilibrium point
- $\omega = 2\pi f = \dfrac{2\pi}{T}$
- $a$–$x$ graph: straight line, gradient $-\omega^2$

## 2 KINEMATICS OF SHM

$$x = A\sin\omega t$$

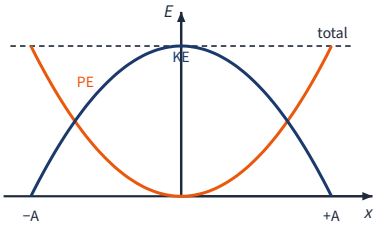
$$v = \omega A\cos\omega t$$

$$v = \pm\omega\sqrt{A^2 - x^2}$$

$$v_{\max} = \omega A \qquad a_{\max} = \omega^2 A$$

$x = A\cos\omega t$ if timing starts at an extreme.

## 3 ENERGY IN SHM




$$E = \tfrac{1}{2}m\omega^2 A^2 = \tfrac{1}{2}kA^2$$

$$E_k = \tfrac{1}{2}m\omega^2(A^2 - x^2)$$

$$E_p = \tfrac{1}{2}m\omega^2 x^2$$

Total energy $\propto$ (amplitude)$^2$; $E_k = E_p$ at $x = A/\sqrt{2}$.

## 4 SYSTEMS IN SHM

### MASS-SPRING

$$T = 2\pi\sqrt{\frac{m}{k}} \qquad \omega = \sqrt{\frac{k}{m}}$$

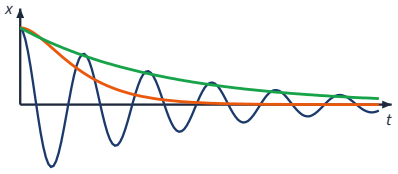
### SIMPLE PENDULUM (SMALL Θ)

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Independent of mass and amplitude; $\theta \lesssim 10^\circ$.

Springs: parallel $k = k_1 + k_2$; series $\dfrac{1}{k} = \dfrac{1}{k_1} + \dfrac{1}{k_2}$

## 5 DAMPED OSCILLATION

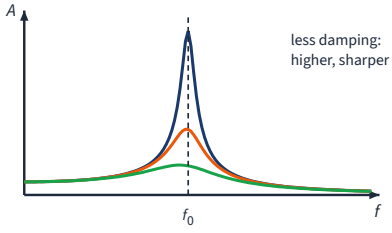



light (navy), critical (orange), heavy (green)

| | |
|---|---|
| **Light** | oscillates, amplitude decays exponentially |
| **Critical** | back to equilibrium fastest, no oscillation |
| **Heavy** | slow return, no oscillation |

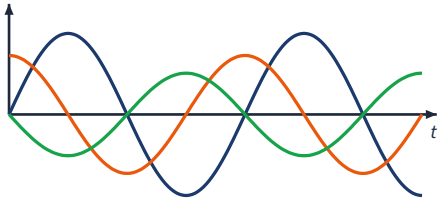
$$A = A_0 e^{-bt}$$

## 6 FORCED OSCILLATION & RESONANCE




- Forced: oscillates at the **driving** frequency
- Resonance: $f_{\text{driver}} = f_0$, maximum amplitude
- More damping: lower, broader peak, shifted slightly to lower $f$

## 7 PHASE RELATIONSHIPS




x (navy), v (orange), a (green)

$v$ leads $x$ by $\pi/2$; $a$ is in antiphase ($\pi$) with $x$.

## 8 COMMON SECTION A TRAPS

- Calculator in degrees for $\sin\omega t$: use **radians**.
- Saying the period depends on amplitude (it does not).
- Pendulum length: to the **centre** of the bob.
- Doubling $A$ doubles $v_{\max}$ but **quadruples** $E$.
- Speed is maximum at $x = 0$, acceleration at $x = \pm A$.

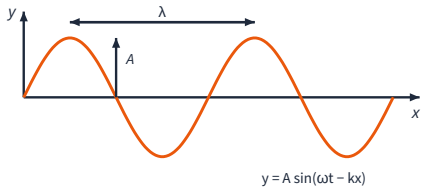
## EXAM TIPS

- ✓Prove SHM by showing $a = -(\text{constant})x$.
- ✓Identify $\omega$ as the coefficient of $t$.
- ✓Use energy for speed at a given $x$.
- ✓Sketch $x$, $v$, $a$ graphs one under another, $T/4$ apart.

# CHAPTER 20 WAVE MOTION

**Key idea:** a progressive wave carries energy without carrying matter; overlapping waves add.

## 1 PROGRESSIVE WAVES




$$v = f\lambda$$

$$y = A\sin(\omega t - kx)$$

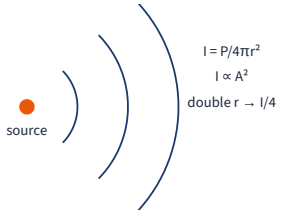
- $k = \dfrac{2\pi}{\lambda}, v = \dfrac{\omega}{k}$
- $+kx$: travels in the $-x$ direction
- Transverse: $\perp$ to travel; longitudinal: $\parallel$

## 2 PHASE DIFFERENCE

$$\phi = \frac{2\pi}{\lambda}\Delta x \qquad \phi = \frac{2\pi}{T}\Delta t$$

In phase: $\Delta x = n\lambda$. Antiphase: $\Delta x = (n + \frac{1}{2})\lambda$.

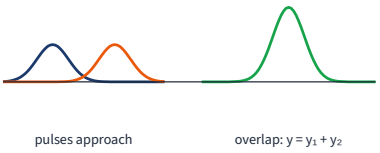
## 3 INTENSITY




$$I = \frac{P}{4\pi r^2} \qquad I \propto A^2$$

Power per unit area, W m$^{-2}$. Point source: $I \propto 1/r^2$ so $A \propto 1/r$.

## 4 SUPERPOSITION




**Constructive** path difference $n\lambda$

**Destructive** path difference $(n + \frac{1}{2})\lambda$

**Coherent:** same frequency, constant phase difference.

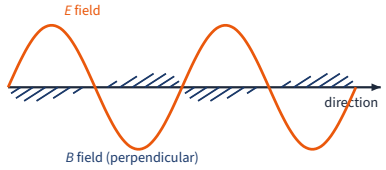
## 5 STANDING WAVES

$$y = 2A\sin kx\ \cos\omega t$$

- Two equal waves travelling in **opposite** directions
- Node to node $= \lambda/2$; node to antinode $= \lambda/4$
- String fixed both ends: $L = \dfrac{n\lambda}{2}$, $f_n = \dfrac{nv}{2L}$

$$v = \sqrt{\frac{T}{\mu}}$$

## 6 ELECTROMAGNETIC WAVES




$$c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \qquad E_0 = cB_0$$

$E \perp B \perp$ direction, in phase. Radio → microwave → IR → visible → UV → X-ray → γ (λ decreasing).

## 7 PROGRESSIVE VS STANDING

| | Progressive | Standing |
|---|---|---|
| Energy | transferred | not transferred |
| Amplitude | same everywhere | zero at nodes, max at antinodes |
| Phase | changes along the wave | same within a loop; antiphase in next loop |
| Wave profile | moves | does not move |

## 8 COMMON SECTION A TRAPS

- Node spacing is $\lambda/2$, not $\lambda$.
- Reading $k$ as the wavelength; $\lambda = 2\pi/k$.
- Doubling the distance quarters $I$ but halves $A$.
- $\mu$ is mass per unit **length** (kg m$^{-1}$).

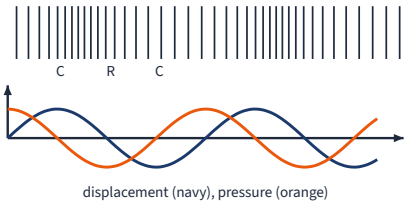
## EXAM TIPS

- ✓From $y = A\sin(\omega t - kx)$ read $A$, $\omega$ and $k$ first.
- ✓Draw the harmonics with N and A labelled.
- ✓Use ratios for the inverse square law.
- ✓State the sources are coherent before discussing interference.

# CHAPTER 21 SOUND WAVES

**Key idea:** sound is a longitudinal pressure wave; air columns and strings resonate at definite frequencies.

## 1 PROPAGATION OF SOUND

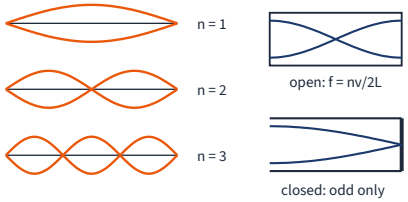


displacement (navy), pressure (orange)

$$v = \sqrt{\frac{\gamma RT}{M}} \qquad v = \sqrt{\frac{E}{\rho}}$$

- Longitudinal: compressions and rarefactions
- Needs a medium; faster in solids
- Gas: $v \propto \sqrt{T}$, independent of pressure

## 2 AIR COLUMNS



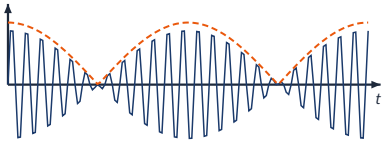

| Pipe | Ends | Harmonics | $f_1$ |
|---|---|---|---|
| Open | A–A | all | $v/2L$ |
| Closed | N–A | odd only | $v/4L$ |

## 3 INTENSITY LEVEL

$$\beta = 10 \log_{10} \frac{I}{I_0}$$

- $I_0 = 1.0 \times 10^{-12}$ W m$^{-2}$ (threshold of hearing)
- +10 dB = intensity ×10; +3 dB ≈ ×2
- $N$ identical sources: $\beta + 10 \log_{10} N$
- Add intensities, **not** decibels

## 4 BEATS



loudness rises and falls $f_1 - f_2$ times per second

$$f_{\text{beat}} = |f_1 - f_2|$$

$$f = \frac{f_1 + f_2}{2}$$

Loudness rises and falls as the waves move in and out of phase.

## 5 DOPPLER EFFECT

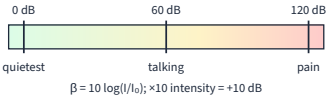
$$f' = f \frac{v \pm v_o}{v \mp v_s}$$

- Upper signs: **approaching** (frequency rises)
- Speeds relative to the air
- Moving reflector: apply twice (observer, then source)

## 6 STRINGS & END CORRECTION

$$f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}}$$

- Node at each fixed end; all harmonics
- Closed pipe: $L + e = \frac{\lambda}{4}$
- Antinode lies slightly **beyond** the open end




## 7 UNIT CHECK

| Quantity | Formula | Unit |
|---|---|---|
| Intensity | $P/4\pi r^2$ | W m$^{-2}$ |
| Intensity level | $10 \log(I/I_0)$ | dB |
| Beat frequency | $\lvert f_1 - f_2 \rvert$ | Hz |
| Mass per length | $\mu$ | kg m$^{-1}$ |

## 8 COMMON SECTION A TRAPS

- Closed pipe has **odd** harmonics only: 1st, 3rd, 5th…
- Adding dB values directly.
- Using °C in $v \propto \sqrt{T}$.
- Doppler: the wrong sign for approach.
- Displacement node = pressure **antinode**.

## EXAM TIPS

- ✓Sketch the air column with N and A before using a formula.
- ✓Convert dB to intensity, add, then convert back.
- ✓Write $f'$ formula with signs explained in words.
- ✓Use $v = 4f(L + e)$ for resonance-tube data.

CHAPTER 22

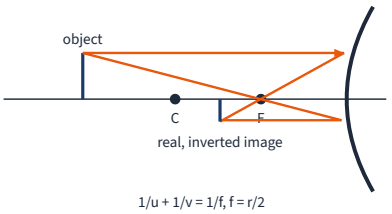
# GEOMETRICAL OPTICS

**Key idea:** mirrors and lenses form images by reflection and refraction; use the real-is-positive convention.

## 1 SIGN CONVENTION

- Real object, real image: **positive**
- Virtual image: $v$ **negative**
- Concave mirror, convex lens: $f > 0$
- Convex mirror, concave lens: $f < 0$
- $m = -\dfrac{v}{u}$: negative = inverted, positive = upright

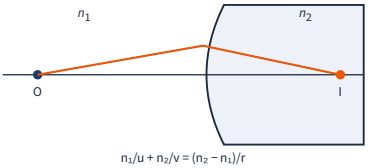
## 2 SPHERICAL MIRRORS




1/u + 1/v = 1/f, f = r/2

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f} \qquad f=\frac{R}{2}$$

## 3 REFRACTION AT A CURVED SURFACE




$n_1/u + n_2/v = (n_2 - n_1)/r$

$$\frac{n_1}{u}+\frac{n_2}{v}=\frac{n_2-n_1}{R}$$

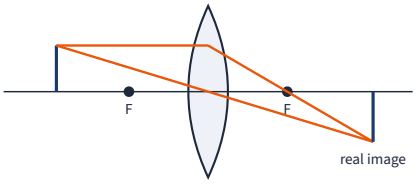
- $R > 0$ if the centre is on the **outgoing** side
- Plane surface: apparent depth $= \dfrac{\text{real depth}}{n}$

## 4 IMAGE NATURE

| Object (converging) | Image |
|---|---|
| $u > 2f$ | real, inverted, diminished |
| $u = 2f$ | real, inverted, same size |
| $f < u < 2f$ | real, inverted, magnified |
| $u < f$ | virtual, upright, magnified |

Convex mirror / concave lens: always virtual, upright, diminished.

## 5 THIN LENSES




1/u + 1/v = 1/f

$$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}+\frac{1}{R_2}\right)$$

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f} \qquad P=\frac{1}{f}$$

Biconvex: both $R$ positive. Power in dioptre (D), $f$ in metres.

## 6 USEFUL RESULTS

- Lenses in contact: $\dfrac{1}{f}=\dfrac{1}{f_1}+\dfrac{1}{f_2}$, $P = P_1 + P_2$
- Same-size real image: $u = v = 2f$
- Minimum object–image distance: $4f$
- Mirror: $f$ does not depend on the medium; lens: $f$ does

## 7 RAY-DIAGRAM RULES

1. Ray parallel to the axis → through (or from) $F$.
2. Ray through the optical centre (lens) or $C$ (mirror) → undeviated / back on itself.
3. Ray through $F$ → emerges parallel to the axis.

## 8 COMMON SECTION A TRAPS

- Dropping the minus sign of a virtual image or a diverging lens.
- Using $f = R$ instead of $R/2$ for a mirror.
- Wrong sign of $R$ in the lensmaker's or surface equation.
- Mixing cm and m when finding power in D.

## EXAM TIPS

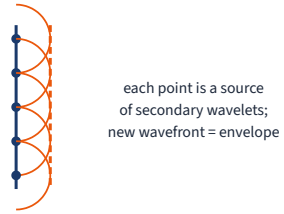
- ✓State the sign convention at the start of the answer.
- ✓Draw a quick ray diagram to check the answer is sensible.
- ✓Use $m = -v/u$ and describe the image in three words.
- ✓Find each image in turn for two-lens systems.

CHAPTER 23

# WAVE OPTICS

**Key idea:** light shows interference, diffraction and polarisation, so it is a transverse wave.

## 1 HUYGENS'S PRINCIPLE

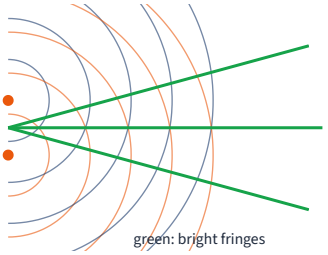



Every point on a wavefront is a source of secondary wavelets; the new wavefront is their **envelope**.

$$n = \frac{c}{v} = \frac{\lambda_0}{\lambda}$$

On refraction $f$ is unchanged; $v$ and $\lambda$ change.

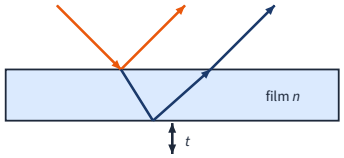
## 2 YOUNG'S DOUBLE SLIT




$$x = \frac{\lambda D}{a} \qquad a \sin\theta = n\lambda$$

- Bright: path difference $n\lambda$; dark: $(n + \frac{1}{2})\lambda$
- White light: white centre, coloured fringes (violet nearest)

## 3 THIN FILMS




path difference 2nt (+ λ/2 for one reflection phase change)

- π phase change on reflection at a **denser** medium
- Film in air (one change): bright $2nt = (m + \frac{1}{2})\lambda$
- Coating on glass (two changes): dark $2nt = (m + \frac{1}{2})\lambda$
- Minimum coating: $t = \dfrac{\lambda}{4n}$
- Air wedge: $x = \dfrac{\lambda}{2\theta}$, dark at contact

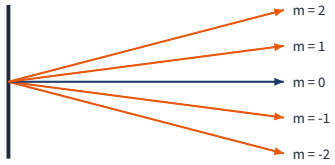
## 4 SINGLE-SLIT DIFFRACTION




$$a \sin\theta = m\lambda \qquad w = \frac{2\lambda D}{a}$$

Minima ($m = 1, 2, \ldots$). Central maximum twice as wide as the others.

## 5 DIFFRACTION GRATING

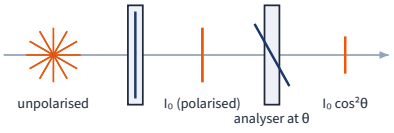



d sin θ = mλ: sharp, bright maxima

$$d \sin\theta = m\lambda \qquad d = \frac{1}{N}$$

- Highest order: largest integer below $d/\lambda$
- More lines: sharper, brighter maxima
- Total maxima $= 2m_{\max} + 1$

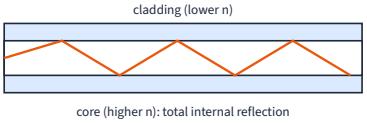
## 6 POLARISATION




$$I = I_0 \cos^2\theta \qquad \tan\theta_B = n$$

- Unpolarised through a polariser: $I_0/2$
- At $\theta_B$: reflected and refracted rays at 90°
- Only **transverse** waves can be polarised

## 7 OPTICAL FIBRES




$$\sin\theta_c = \frac{n_2}{n_1} \qquad \mathrm{NA} = \sqrt{n_1^2 - n_2^2}$$

Total internal reflection: $n_{\text{core}} > n_{\text{cladding}}$, angle > $\theta_c$. Modal dispersion spreads pulses; attenuation $= \dfrac{10}{L} \log \dfrac{P_{\text{in}}}{P_{\text{out}}}$ dB per unit length.

## 8 COMMON SECTION A TRAPS

- Mixing $a$ (slit separation or width) with $d$ (grating spacing).
- Forgetting the extra λ/2 from a reflection at a denser medium.
- Lines per mm → lines per m before $d = 1/N$.
- Using Malus's law on **unpolarised** light.
- Single slit: $a \sin\theta = m\lambda$ gives **minima**.

## EXAM TIPS

- ✓State "coherent sources" and the path-difference condition.
- ✓For thin films, count the phase changes first.
- ✓Check $\sin\theta \leq 1$ for the highest grating order.
- ✓Fringe spacing ∝ λ: red fringes are wider than blue.

CHAPTER 24

# QUANTUM PHYSICS

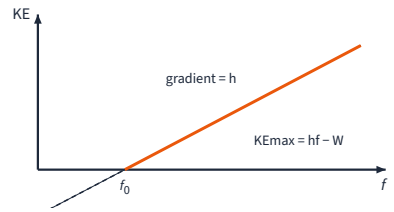
**Key idea:** light comes in photons of energy $hf$, and moving particles behave as waves of wavelength $h/p$.

## 1 PHOTON

$$E = hf = \frac{hc}{\lambda} \qquad p = \frac{h}{\lambda}$$

- A quantum (packet) of electromagnetic energy
- 1 eV $= 1.60 \times 10^{-19}$ J
- $E$ (eV) ≈ 1240 ÷ λ (nm)

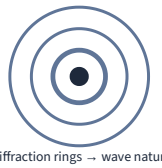
## 2 PHOTOELECTRIC EFFECT




$$hf = \phi + K_{\max} \qquad eV_s = K_{\max}$$

Graph of $K_{\max}$ against $f$: gradient $h$, $f$-intercept $f_0 = \phi/h$.

## 3 WAVE-PARTICLE DUALITY



electron diffraction rings → wave nature, λ = h/p

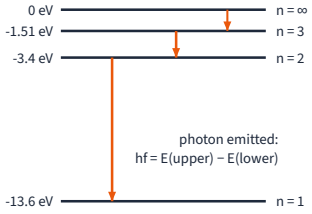
$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2meV}}$$

Electron diffraction rings show the wave nature of electrons.

## 4 WAVE THEORY FAILS

1. There is a **threshold frequency** $f_0$.
2. $K_{\max}$ depends on frequency, **not** intensity.
3. Emission is **instantaneous**.
4. Intensity changes only the number of photoelectrons.

## 5 ATOMIC STRUCTURE

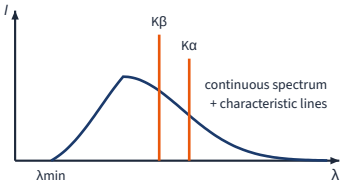



$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

$$hf = E_i - E_f$$

Lyman → $n = 1$ (UV); Balmer → $n = 2$ (visible); Paschen → $n = 3$ (IR). Bohr: $mvr = \dfrac{nh}{2\pi}$.

## 6 X-RAYS




$$\lambda_{\min} = \frac{hc}{eV} \qquad 2d\sin\theta = n\lambda$$

- Continuous spectrum: braking (bremsstrahlung)
- Sharp lines: characteristic of the target
- Bragg angle $\theta$ is measured from the **planes**

## 7 NANOSCIENCE

- Size 1–100 nm; huge surface-area-to-volume ratio ($3/r$ for a sphere)
- Quantum confinement: $E_n = \dfrac{n^2h^2}{8mL^2}$, smaller dot → bluer light
- STM: tunnelling current depends exponentially on the gap

## 8 COMMON SECTION A TRAPS

- Using eV in $hf = \phi + K$ while $h$ is in J s.
- Saying brighter light gives faster photoelectrons.
- Forgetting energy levels are **negative**.
- Using the electron mass for a proton in $\lambda = h/p$.
- $\lambda_{\min}$ depends on the voltage, not the target.

## EXAM TIPS

- ✓ Convert every energy to joules (or every one to eV) before substituting.
- ✓ Draw the transition arrow on the level diagram.
- ✓ For $\lambda_{\min}$, all the electron KE becomes one photon.
- ✓ Quote $h = 6.63 \times 10^{-34}$ J s.

# CHAPTER 25 NUCLEAR PHYSICS

**Key idea:** the mass defect of a nucleus is its binding energy; unstable nuclei decay randomly.

## 1 THE NUCLEUS

$${}^{A}_{Z}\mathrm{X}$$

- $Z$: protons; $A$: nucleons; $N = A - Z$
- Isotopes: same $Z$, different $A$
- $R = R_0 A^{1/3}$, $R_0 \approx 1.2$ fm
- Nuclear density the same for all nuclei

## 2 MASS DEFECT & BINDING ENERGY

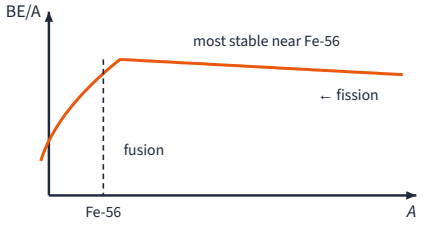
$$\Delta m = Z m_p + (A - Z) m_n - m_{\text{nucleus}}$$

$$B = \Delta m\, c^2$$

$$1\ \text{u} = 931.5\ \text{MeV}$$

Binding energy: energy needed to separate a nucleus into its nucleons.

## 3 BINDING ENERGY PER NUCLEON



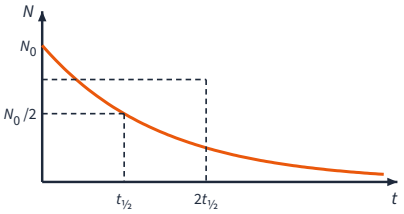

- Peak ≈ 8.8 MeV near $^{56}$Fe: most stable
- Light nuclei → **fusion**; heavy nuclei → **fission**

## 4 RADIOACTIVE DECAY

| Emission | Change |
|---|---|
| α ($^4_2$He) | $A - 4$, $Z - 2$ |
| $\beta^-$ | $A$ same, $Z + 1$ |
| γ | no change |

- **Spontaneous**: not affected by conditions
- **Random**: cannot predict which nucleus decays next

## 5 DECAY LAW



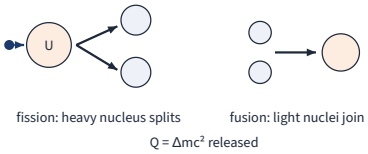

$$N = N_0 e^{-\lambda t} \qquad A = \lambda N$$

$$t_{1/2} = \frac{\ln 2}{\lambda}$$

$$N = N_0 \left(\tfrac{1}{2}\right)^{t/t_{1/2}}$$

$\ln A$ against $t$: straight line, gradient $-\lambda$. Activity in Bq.

## 6 NUCLEAR REACTIONS




$$Q = \left(\sum m_{\text{reactants}} - \sum m_{\text{products}}\right) c^2$$

Conserved: $A$, $Z$ (charge), energy, momentum. $Q > 0$: energy released.

## 7 FISSION VS FUSION

| | Fission | Fusion |
|---|---|---|
| Process | heavy nucleus splits (e.g. $^{235}$U + n) | light nuclei join (e.g. $^2$H + $^3$H) |
| Needs | slow neutron; chain reaction | very high temperature |
| Energy | ≈ 200 MeV per fission | more per kg of fuel |
| Why | products have higher B/A | products have higher B/A |

## 8 COMMON SECTION A TRAPS

- Using total binding energy instead of **per nucleon** for stability.
- Forgetting electron masses when using atomic masses.
- Taking $t_{1/2} = \lambda \ln 2$ (it is $\ln 2 / \lambda$).
- Mixing u, kg and MeV in one line.
- Time in years with $\lambda$ in $\text{s}^{-1}$.

## EXAM TIPS

- ✓Balance $A$ and $Z$ before finding $Q$.
- ✓Use 1 u = 931.5 MeV to skip $c^2$.
- ✓Count half-lives when $t$ is a whole multiple of $t_{1/2}$.
- ✓Answer "why energy is released" with binding energy per nucleon.